# 31338 Network Servers

## Lab 6b: Configuring and Prioritising DNS clients

## Aims

1. Configure a DNS client in CentOS
2. Configure a DNS client in Windows

## Task 1: Configuring DNS clients in CentOS 10

In Linux, the DNS client software is usually called a "resolver". The configuration file for the resolver is located at */etc/resolv.conf*. Remind that in the static network configuration, two properties *dns* and *dns-search* are set in the file ens192.nmconnection.

The `dns` the IP address of a name server that this client should attempt to connect to. There could be several lines of name server. Normally you would put your closest nameserver first. For example, if your resolv.conf file contained the lines

```
nameserver 138.25.9.1

nameserver 138.25.8.1
```

then the resolver would first try and use the name server at IP address 138.25.9.1. If that server is unavailable, it would try 138.25.8.1.

The `*dns-search*` specifies a list of domain names to be searched when an unqualified hostname is used in a lookup query. For example, if resolv.conf contained the line:

```
search it.uts.edu.au iwork.uts.edu.au
```

and a user typed in "ping ns", then first the resolver library would try and find an IP address for ns.it.uts.edu.au. If unsuccessful, it would try to find an IP address for ns.iwork.uts.edu.au. If still unsuccessful, it would return a "host not found" error to the user.

The Linux system automatically generate resolv.conf based on the network configurations. So rather than editing resolv.conf directly to change them, we need to modify the network configurations.

First, have a look at the file */etc/resolv.conf.* What are the name servers and dns-search domains?

Then, run `ping ns`. Where is the response from?

Now, open the network configuration file ens192.nmconnection in a text editor. Change the values of *dns* and *dns-search*. In both two settings, you can specify more than one values using semi-colon to separate them, like

```
dns=10.0.2.1;10.0.2.2;
dns-search=it.netserv.edu.au;netserv.edu.au;
```

You can prioritise the DNS on the current network adaptor using

```
dns-priority=10
```

Reload and restart the network interface ens192, and check /etc/resolv.conf. Is the order of search domain and name server changed? Which name server will be searched first?

# Task 2: Configuring DNS clients in Windows

In Windows Server you can override the DNS settings by opening the Network Connections panel from *Server Manager -> Local Server -> Ethernet1*. Then right-click on the "Ethernet1" adapter and choose *Properties -> Internet Protocol Version 4 (TCP/IPv4) -> Properties*. Set the *Preferred DNS Server* to 10.0.2.2. To the search domain, click *Advacend…*, go to the *DNS* tab, choose *Append these DNS suffixes (in order)*, then add `netserv.edu.au` and `it.netserv.edu.au`.

The priority of nameserver can be checked through PowerShell. Open a PowerShell terminal and run the following command

```
Get-NetIPInterface
```

Look at the column *InterfaceMetric*. Are the priority values of Ethernet0 and Ethernet1 the same?

If we would like to prioritise the second adaptor Ethernet1, go back to the *Advanced TCP/IP Settings* in *Ethernet1 Properties*. Uncheck *Automatic metric*, then set a new value in *Interface metric*.

You can verify the DNS server search order with the following PowerShell command:

```
Get-DNSClientServerAddress
```

Most likely you will find that the DNS server configured on Ethernet0 has priority over the DNS server configured on Ethernet1.

# Task 3: Testing The Resolver

On both CentOS and Windows, try the ping command without the DNS suffix. Are the response from each other?

On CentOS, use the dig command without localhost, i.e.,

```
dig ns.it.netserv.edu.au
```

What is the answer?

On Windows, what is the IP address when type *nslookup*?

### Remember to take a snapshot of your virtual machines!